Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *ARABICMMLU* | ***Context:*** *MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is the strongest dimension for this deployment. Both the benchmark and deployment use single-token MCQ accuracy [Q50, Q53], and the elicitation explicitly confirmed that for MCQ tasks 'explanations are not graded, so any correct explanation adequately supports the student's understanding without exam risk' [elicitation A3]. The benchmark provides multiple prompt-language and output-alphabet configurations [Q47] with detailed per-configuration results [Q105, Q106, Q108, Q110], offering rich diagnostic capacity. Calibration analysis (r > 0.9 between answer probability and accuracy [Q80]) provides additional signal quality. Models are evaluated in zero-shot and few-shot settings across 35 models [Q45, Q46], giving comparative reference points. The remaining concerns are residual: the form does not capture explanation quality (which is acceptable per elicitation A3) and does not evaluate generative tutoring output that the deployment may also produce. Literacy is a non-issue since target users are enrolled students [regional_context literacy section].

ID	Evidence
Q45	“Our experiments focus on zero-shot and few-shot settings across 35 models.” (p.5)
Q46	“This includes 22 open-source multilingual models (XGLM (Lin et al., 2022), BLOOMZ (Muennighoff et al., 2022), mT0 (Muennighoff et al., 2022), Falcon (Penedo et al., 2023), and LLaMA2 (Touvron et al., 2023), across various sizes), 11 open-source Arabic-centric models (AraT5 (Nagoudi et al., 2022), AraGPT2 (Antoun et al., 2021b), AceGPT (Huang et al., 2023) and Jais (Sengupta et al., 2023), also across various sizes), and 2 closed-source models (GPT-3.5: gpt-3.5-turbo (Ouyang et al., 2022) and GPT-4: gpt-4-0613 (OpenAI, 2023)).” (p.5)
Q47	“We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output.” (p.5)
Q50	“Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options.” (p.6)
Q53	“For closed-source models, we determine the answer based on the first token generated in the text using a regular expression.” (p.6)
Q80	“In Figure 8, we observe that the three open-source models are well calibrated with correlation scores r > 0.9.” (p.8)
Q105	“Zero-shot LLM performance (% accuracy) with Arabic prompt and Arabic alphabetic output, combined across subject groups.” (p.16)
Q106	“"Average" means the average across all questions in ArabicMMLU.” (p.16)
Q108	“Zero-shot LLM performance (% accuracy) with Arabic prompt and English alphabetic output, combined across subject groups.” (p.17)
Q110	“Zero-shot LLM performance (% accuracy) with English prompt and Arabic alphabetic output, combined across subject groups.” (p.18)

Strengths

- Single-token MCQ accuracy directly matches deployment's MCQ scoring needs [Q50, Q53]
- Four prompt-language/output-alphabet configurations tested, with per-configuration results published [Q47, Q105, Q106, Q108, Q110]
- Calibration analysis (r > 0.9) confirms answer-probability reliability for open-source models [Q80]
- Per-subject and per-level result breakdowns enable diagnostic targeting [Q102, Q103]

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Q47	“We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output.” (p.5)
Q50	“Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options.” (p.6)
Q53	“For closed-source models, we determine the answer based on the first token generated in the text using a regular expression.” (p.6)
Q80	“In Figure 8, we observe that the three open-source models are well calibrated with correlation scores r > 0.9.” (p.8)
Q102	“Table 8 presents the detailed zero-shot results across subjects and education levels, while Table 9, Table 10, Table 11 display the results with different prompts and alphabetic outputs (complementing the main result at Table 8).” (p.15)
Q103	“Zero-shot LLM performance (% accuracy) with English prompt and English alphabetic output, for each subject and education level.” (p.15)
Q105	“Zero-shot LLM performance (% accuracy) with Arabic prompt and Arabic alphabetic output, combined across subject groups.” (p.16)
Q106	“"Average" means the average across all questions in ArabicMMLU.” (p.16)
Q108	“Zero-shot LLM performance (% accuracy) with Arabic prompt and English alphabetic output, combined across subject groups.” (p.17)
Q110	“Zero-shot LLM performance (% accuracy) with English prompt and Arabic alphabetic output, combined across subject groups.” (p.18)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Expected output modality (single-letter MCQ answer) matches the deployment's primary MCQ tutoring use case [Q50, Q53; elicitation A3]. The deployment may also need free-form explanations, which the benchmark does not evaluate — but the elicitation confirmed this is non-critical for MCQ contexts [elicitation A3].

OF-2: Text-to-speech is not required by the deployment (text-based interface per regional_context platform_context). Not applicable.

OF-3: Target population is enrolled students with high literacy by definition [literacy_and_educational_attainment note]; mobile-dominant access is documented [WEB-11]. Output form is compatible with regional access patterns.

OF-4: Documented form mismatches are minimal: (a) explanation-quality output is not evaluated, but elicitation confirmed this is acceptable [elicitation A3]; (b) generative tutoring output (e.g., step-by-step explanations) falls outside benchmark scope but is a deployment use case the benchmark does not validate.

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Q45	“Our experiments focus on zero-shot and few-shot settings across 35 models.” (p.5)
Q50	“Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options.” (p.6)
Q53	“For closed-source models, we determine the answer based on the first token generated in the text using a regular expression.” (p.6)
WEB-8	~100% internet penetration in UAE/KSA/Kuwait — text-based output reaches target users (https://statisticsoftheworld.com/ranking/internet-users)

WEB-11	83% mobile ownership in Arab States; mobile-dominant text delivery is feasible (https://www.itu.int/dms_pub/itu-d/opb/ind/D-IND-SDDT_ARB-2025-PDF-E.pdf)